

AcousticDiffusion: Semantically Conditioned Audio-Guided Diffusion Policy for Search-and-Rescue Assistance

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Abstract—Navigating toward human callers is an important capability for rescue robots operating where visual contact is degraded or occluded. We present AcousticDiffusion, a semantically conditioned, audio-guided diffusion policy for human-directed navigation. A frozen pretrained audio recognizer processes 10.24s windows, with speech gating and distress-aware prioritization converting recognition outputs into source-level navigation roles. Microphone-array direction-of-arrival measurements are recursively integrated into a robot-centric Bayesian bird’s-eye-view belief field. Ego-motion compensation aligns successive observations, progressively constraining source position while preserving bearing-induced range uncertainty. The semantic belief, recent acoustic observations, audio features, and robot state condition a diffusion model that generates waypoint trajectories. On a synthetic-navigation validation set using recorded audio, AcousticDiffusion achieves a mean end-point bearing error of 11.20° , with 91.78% of trajectories aligned within 30° of the caller. Distractor rejection ranges from 89.20% to 98.99%, and the policy favors a HELP-designated caller over a competing speaker in 91.07% of windows. Deployed online on a ZSL-1 quadruped without additional retraining, it achieves a mean bearing error of 64.9° , compared with 98.2° for A* and 90.4° for RRT, with a mean planner compute time of 6.07 ms. Despite imperfect acoustic localization, the reported mean final source distance is reduced from 3.96 m for the classical planners using ODAS-derived (Open embedded Audition System) guidance to 2.48 m, a 37.4% improvement. These results demonstrate the framework’s ability to translate uncertain acoustic observations into closer approaches to human callers.

I. INTRODUCTION

Navigating toward a sound source is a natural and practical capability for service and rescue robots operating in human environments. Spoken commands, ringing phones, industrial alarms, and distress calls can reveal events and people that are outside the robot’s field of view or occluded by obstacles. Unlike purely geometric acoustic navigation, however, rescue missions require the robot to determine not only *where* a sound originates, but also *which* sound should guide its motion. A person calling or screaming for help may coexist with speech, machinery, alarms, reflections, and ambient noise. The navigation policy must therefore recognize the semantic class of each acoustic observation, select the task-relevant source, and generate a safe trajectory toward it.

Sound-source navigation is further complicated by the bearing-only nature of microphone-array observations. A

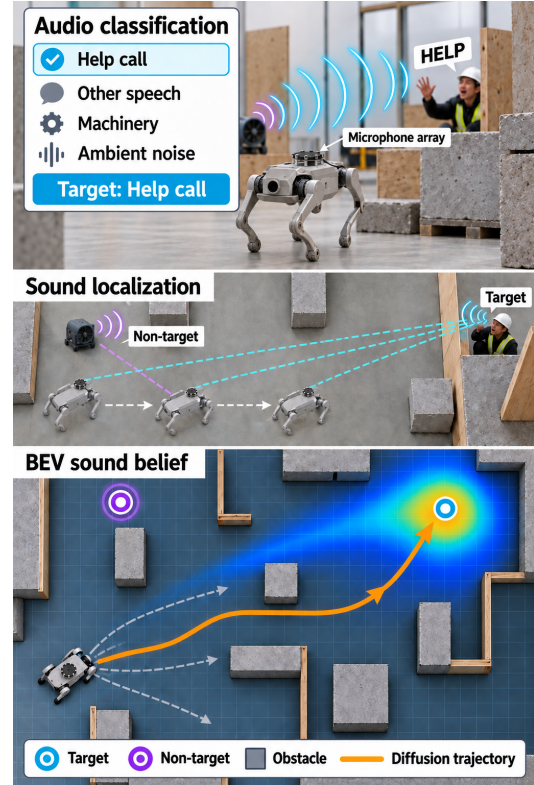


Fig. 1: Overview of ACOUSTICDIFFUSION. The quadruped robot identifies a task-relevant acoustic event among competing sound classes, accumulates direction-of-arrival observations into a bird’s-eye-view sound belief, and generates a diffusion-based trajectory toward the target while avoiding obstacles.

single Direction-of-Arrival (DoA) measurement constrains the source to a ray in free space rather than to a point, providing no direct estimate of range. Recovering source position requires observations from spatially separated robot poses or a strong geometric prior. Robot motion is therefore not only a means of reaching the source, but also an information-gathering action: changes in viewpoint create acoustic parallax, allowing successive bearing measurements to triangulate the source and reduce localization uncertainty.

Classical systems typically address this problem using an explicit acoustic localization front end, such as GCC-PHAT [1], SRP-PHAT [2], or ODAS [3], followed by a separate navigation planner. Although this modular design is interpretable, localization and control are treated as largely independent problems. The planner generally acts on a point estimate after localization, rather than on the evolving spatial

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uncertainty, and cannot directly use motion to resolve bearing ambiguity. Moreover, geometric localization alone cannot determine whether a detected source corresponds to a distress call, irrelevant speech, machinery, or background noise.

Recent learning-based approaches have demonstrated the value of coupling acoustic perception with embodied navigation [4], [5], [6], [7]. These methods show that audio can provide navigation cues beyond the robot’s visual field of view and that acoustic observations accumulated during motion can improve goal-directed navigation. However, most established benchmarks operate primarily in simulation using rendered monaural or binaural audio and formulate the task as reaching a designated sound-emitting goal. More recent work has investigated microphone-array-based navigation on physical robots [8] and sim-to-real transfer of audio-visual navigation policies [9]. Nevertheless, jointly selecting a task-relevant sound based on its semantic meaning, recursively resolving bearing-only localization uncertainty through robot motion, and directly conditioning trajectory generation on the resulting acoustic belief remain largely unexplored.

In parallel, diffusion models have emerged as expressive policies for sequential robot control, including contact-rich manipulation [10] and real-world mobile navigation [11]. By representing a distribution over future actions or waypoints, diffusion policies are particularly well suited to navigation problems in which multiple feasible trajectories may satisfy the same perceptual objective.

We introduce **AcousticDiffusion**, a semantically conditioned, audio-guided diffusion policy for rescue navigation, including visually occluded environments. A frozen AudioSet-pretrained Audio Spectrogram Transformer (AST) processes 10.24 s audio windows. Speech gating and distress-aware speaker prioritization convert its outputs into source-level navigation roles, retaining ordinary speech as a potential target. These semantics are integrated with microphone-array DoA measurements in a robot-centric Bayesian bird’s-eye-view (BEV) field. Ego-motion compensation aligns evidence from successive viewpoints, allowing bearing-only observations to progressively constrain source position while preserving localization uncertainty. The semantic BEV representation, recent DoA and role tokens, audio features, and robot state jointly condition an ε -prediction diffusion model that generates target-directed waypoint trajectories. The main contributions of this work are:

- 1) A **semantically conditioned diffusion navigation framework** that adapts a frozen AudioSet-pretrained AST to source-level navigation through speech gating and distress-aware speaker prioritization. Rather than directly using predicted sound labels, this adaptation converts recognition scores into navigation roles while retaining ordinary speech as a potential target. The resulting semantic conditioning, Bayesian BEV belief, DoA tokens, and ego-state guide an ε -prediction diffusion policy to generate target-directed waypoint trajectories.
- 2) A **recursive semantic Bayesian BEV sound field** that accumulates bearing-only DoA evidence with

ego-motion compensation and associates spatial evidence with source semantics. Successive observations progressively constrain source position, while spatial belief and uncertainty readouts condition trajectory generation. Conditioning ablations identify the BEV role maps as the principal semantic pathway supporting target alignment.

- 3) A **ROS 2 implementation and real-robot evaluation** on a ZSL-1 quadruped, integrating microphone-array observations, semantic audio processing, odometry, diffusion trajectory generation, and LiDAR/IMU based collision avoidance. Experiments assess navigation toward task-relevant sound sources amid competing sounds and physical obstacles. The ROS 2 implementation and trained model checkpoints will be released upon acceptance.

II. RELATED WORK

A. Acoustic Localization and Bearing-Only Tracking

Classical microphone-array localization methods include GCC-PHAT [1], which estimates inter-channel time delays, and SRP-PHAT [2], which evaluates steered responses over candidate source locations. ODAS [3] integrates localization, tracking, and separation for real-time robot audition and supplies our DoA observations.

A single bearing does not determine source range; localization requires additional geometric constraints, such as observations from spatially separated robot poses [12]. Particle filters [13] provide one approach to nonlinear, non-Gaussian tracking. AcousticDiffusion instead accumulates bearing evidence in a grid-based log-odds field, using odometry to align successive observations in the robot frame. This preserves multiple spatial hypotheses and localization uncertainty for conditioning the trajectory policy.

B. Audio-Guided and Audio-Visual Navigation

Learning-based audio-guided navigation has received growing attention in embodied AI and robotics. Gan et al. [4] formulated audio-visual embodied navigation as the task of reaching a sound source from egocentric auditory and visual observations. SoundSpaces [5] subsequently provided realistic acoustic simulation in scanned 3-D environments and established a widely used AudioGoal navigation benchmark. Chen et al. [6] further introduced learned waypoint prediction together with a spatially grounded acoustic memory, allowing an agent to accumulate auditory information as it moves rather than acting only on the current observation. SoundSpaces 2.0 [14] extended this line of work with continuous, geometry-based acoustic simulation and configurable acoustic sensing.

Beyond navigation toward continuously active sound sources, subsequent work has considered richer acoustic settings. Semantic Audio-Visual Navigation [7] introduced semantically grounded and temporally intermittent acoustic events, together with persistent multimodal memory that permits navigation after the sound has ceased. Move2Hear [15] demonstrated that agent motion can also be selected actively

to improve separation of a target sound from competing sources. Sound Adversarial Audio-Visual Navigation [16] studied navigation in the presence of interfering acoustic sources, while Younes et al. [17] considered moving targets, distractor sounds, and previously unheard acoustic events. These works demonstrate the importance of semantic reasoning, acoustic memory, and active perception when navigation takes place in complex sound scenes.

Most of these approaches are developed and evaluated primarily using simulated binaural observations. Work closer to physical robot audition includes Rao and Chowdhury [8], who demonstrated audio-aware navigation with a microphone-array-equipped mobile robot in indoor environments. Chen et al. [9] subsequently studied sim-to-real transfer for audio-visual navigation through frequency-adaptive acoustic-field prediction and demonstrated navigation to sounding objects on a physical robot. These studies establish the feasibility of real-world audio-guided navigation, but do not jointly address semantic selection of a task-relevant source, recursive bearing-only localisation with explicit uncertainty, and generative trajectory prediction. AcousticDiffusion combines these elements by associating source semantics with tracked DoA observations, accumulating the selected bearing evidence into a robot-centric Bayesian belief, and conditioning trajectory generation directly on that evolving belief.

C. Diffusion Policies for Robot Navigation

Diffusion Policy [10] and diffusion-based imitation learning [18] generate multimodal action sequences through conditional denoising. For mobile robots, NoMaD [11] combines visually guided navigation and goal-agnostic exploration. When coupled with person detection or a suitable visual goal, visually conditioned trajectory generation could support approaching people for assistance; however, direct visual localization requires visibility. AcousticDiffusion instead uses acoustic semantics and a recursive spatial belief derived from bearing-only observations and ego-motion, providing guidance toward human callers even when visual contact is unavailable.

III. METHOD

AcousticDiffusion generates trajectories toward human sound sources using frozen audio recognition, a recursive Bayesian bird’s-eye-view (BEV) field, and a conditional diffusion policy (Fig. 2). The system receives ODAS-separated audio, source-bearing observations, and robot odometry. Speech determines whether a source attracts the robot; distress provides a preference among speakers rather than a prerequisite for approach. The default configuration uses acoustic conditioning without a visual encoder. Recognizer parameters remain fixed while the trajectory policy is trained.

A. Frozen Speech Recognition and Speaker Preference

a) Audio input and pretrained representation.

Each tracked source is represented by a trailing 10.24 s audio window sampled at 16 kHz. Windows advance every

0.5 s. The window duration and decision thresholds are loaded from the recognizer checkpoint, ensuring consistent offline and online preprocessing. Audio is downmixed, resampled, DC-corrected, adjusted to the prescribed duration, and RMS-normalized using the shared canonicalization pipeline. Shorter clips are padded; padding does not replace the information supplied by real audio. A 128-bin log-mel filterbank represents logarithmic sound energy over time across mel-frequency bands, using AST’s 1024-frame input budget.

An AudioSet-pretrained Audio Spectrogram Transformer (AST) extracts acoustic features, with attention pooling providing a 128-dimensional latent $\mathbf{z}_{t,i}$ for source i at time t . The backbone is frozen and no task-specific speech-recognizer training is performed in the current pipeline. Its pretrained AudioSet labels supply the speech score $s_{t,i} \in [0, 1]$ directly, one scalar per tracked source per time step. The previous 13-class classifier and separate NONE/HAIL/HELP intent head are not constructed in the speech-first configuration.

b) Speech gating and distress preference.

The system distinguishes other sounds, neutral speech, and distress calls. These outputs arise from two decisions with different roles: speech detection gates attraction, whereas distress ranks speech sources. The current checkpoint additionally contains an already-fitted distress probe, held fixed during policy training and inference. Speech gating is therefore zero-shot, while distress ranking uses fixed, previously fitted parameters.

For a detected speaker, distress modulates preference within a bounded interval $[\beta, 1]$, where $\beta = 0.6$ in the documented configuration. A neutral speaker therefore retains a nonzero attraction preference; missing distress lowers priority rather than converting the person into an obstacle. Sources are compared using speech evidence in logit space so that the weaker distress preference does not override the speech gate. Source roles are ATTRACT or NEUTRAL; no source receives an AVOID role. Calibrated thresholds are restored from the checkpoint rather than replaced by a generic threshold of 0.5.

B. Recursive Bayesian Field and BEV Estimation

a) Bearing evidence and temporal alignment.

A single direction-of-arrival (DoA) observation constrains source direction but not range. For a BEV cell $\mathbf{q} = (x, y)$ and bearing $\theta_{t,i}$, the angular likelihood is represented by a von Mises ridge:

$$\begin{aligned} \phi(\mathbf{q}) &= \text{atan2}(y, x), \\ \ell_{t,i}(\mathbf{q}) &\propto \exp\{\kappa_{t,i}[\cos(\phi(\mathbf{q}) - \theta_{t,i}) - 1]\}, \end{aligned} \quad (1)$$

where $\phi(\mathbf{q})$ is the bearing of cell $\mathbf{q}$ as seen from the robot’s own origin, i.e. the azimuth a source located at $\mathbf{q}$ would produce; $\theta_{t,i}$ is the measured DoA bearing for tracked source i at time t ; and $\ell_{t,i}(\mathbf{q})$ is the resulting angular likelihood assigned to cell $\mathbf{q}$, a ridge peaked where $\phi(\mathbf{q}) = \theta_{t,i}$ and uniform in range. The concentration $\kappa_{t,i}$ depends on detection energy or activity (defined below); larger $\kappa_{t,i}$ sharpens the

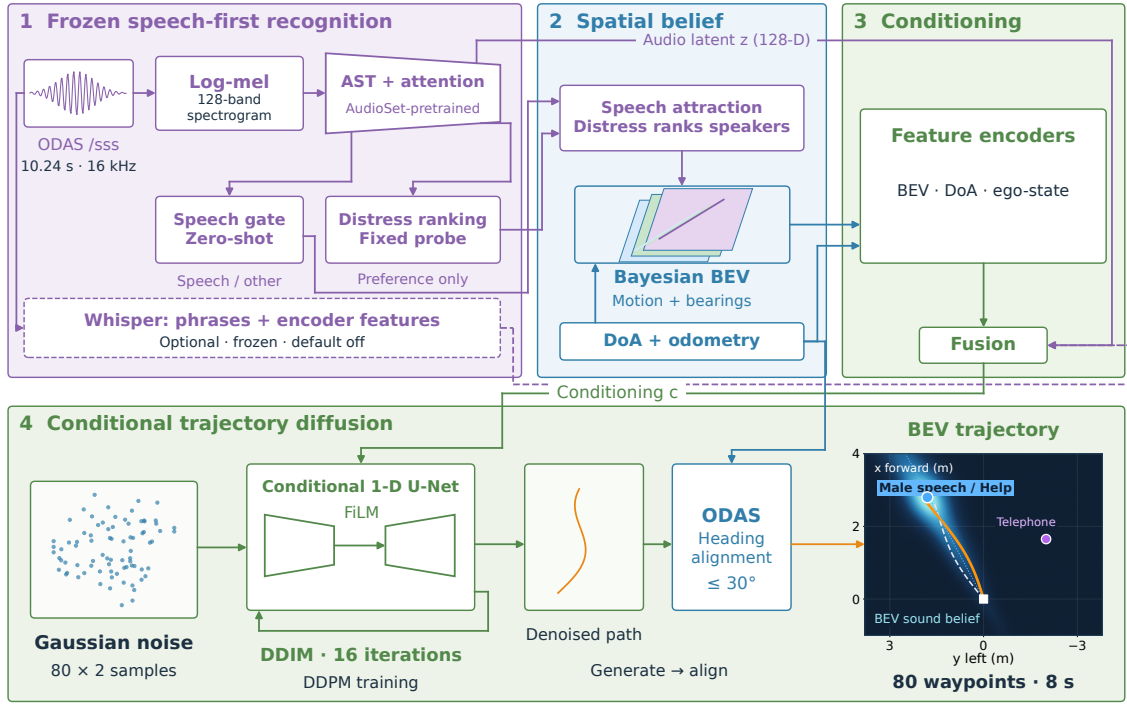


Fig. 2: Overview of AcousticDiffusion. Pretrained audio recognition provides semantic predictions and audio features. Sound-source bearings and odometry update a Bayesian BEV belief. Fused conditioning guides a 1-D U-Net that generates trajectories from Gaussian noise through 16 DDIM iterations. The BEV output and denoising stages are illustrative.

ridge around $\theta_{t,i}$, while $\kappa_{t,i} \rightarrow 0$ flattens it toward a uniform prior over bearing.

The concentration $\kappa_{t,i}$ depends on detection energy or activity. For a fixed direction, this observation model is uniform in range and does not manufacture a distance estimate from a single bearing.

The field accumulates evidence in robot-centric log-odds grids. Before incorporating current observations, the previous field is warped using the odometric pose increment and attenuated by temporal decay:

$$L_t^- = \rho_t \mathcal{W}_{\Delta \mathbf{p}_t}(L_{t-1}), \quad (2)$$

where L_{t-1} is the accumulated log-odds field from the previous time step and L_t^- is its predicted value at time t , before that step's new bearing evidence is folded in. $\Delta \mathbf{p}_t$ is the odometric pose increment (translation and rotation) between $t-1$ and t , and $\mathcal{W}_{\Delta \mathbf{p}_t}(\cdot)$ is the rigid warp operator that re-expresses the previous field in the robot's new body frame under that motion, so that cells still refer to the same world locations after the robot has moved. $\rho_t \in (0, 1]$ is a scalar temporal-decay (forgetting) factor applied at time t , implementing the field's evidence half-life: $\rho_t < 1$ lets stale evidence fade so that a source which has moved does not leave a persistent ghost.

New angular evidence updates the aligned grids. Observations from spatially separated poses can then intersect, progressively constraining source range when robot motion provides sufficient parallax. Decay limits the persistence of obsolete evidence.

b) Semantic field construction.

Untracked SSL potentials update the agnostic field without assigned semantic roles. Tracked SST bearings are associated with audio-derived roles through source identifiers and contribute to the semantic field. The speech-first configuration supplies attraction and neutral evidence without avoidance assignments. Inactive tracks and near-vertical detections are filtered while preserving the association between each bearing and its semantic information.

The online runner buffers ODAS callbacks and assembles observations at 10 Hz, using at most the latest observation for each track in a tick. Odometry history is sampled on the same time base. Shared dataset and online window construction preserves BEV-channel layout, token ordering, pose conventions, and field-readout dimensions.

c) Spatial estimates.

The policy receives accumulated BEV evidence and recent observation history, together with source-position and uncertainty readouts. A centroid estimate can be obtained by normalizing the log-odds field:

$$\pi_t(\mathbf{q}) = \frac{\exp L_t(\mathbf{q})}{\sum_{\mathbf{q}'} \exp L_t(\mathbf{q}')}, \quad \hat{\mathbf{s}}_t = \sum_{\mathbf{q}} \pi_t(\mathbf{q}) \mathbf{q}. \quad (3)$$

$L_t(\mathbf{q})$ is the fully updated log-odds field at cell $\mathbf{q}$ and time t (i.e. L_t^- after that step's bearing evidence has been incorporated). $\pi_t(\mathbf{q})$ is the corresponding posterior probability mass at cell $\mathbf{q}$, obtained by a softmax over the log-odds field, so that π_t is a proper probability distribution over the grid ($\sum_{\mathbf{q}} \pi_t(\mathbf{q}) = 1$); $\mathbf{q}'$ is a dummy variable ranging over all grid cells, used only for that normalizing sum. $\hat{\mathbf{s}}_t$ is the

resulting centroid estimate of the source position at time t : the probability-weighted mean grid coordinate.

Normalizing log-odds retains spatial contrast that may be lost after sigmoid saturation. Spread and confidence accompany the spatial readout. The field estimate remains distinct from the learned source estimate produced by the policy's auxiliary head.

C. Conditional Trajectory Generation

a) Policy conditioning.

A CNN encodes the BEV belief with spatial pooling. A DoA encoder embeds source tokens and role information using masked mean and maximum aggregation followed by temporal processing. An ego-state encoder processes pose history and field readouts. These features and the pooled audio latent form the conditioning vector:

$$\mathbf{c}_t = f_{\text{fuse}}(f_{\text{BEV}}(B_t), f_{\text{DoA}}(D_t), f_{\text{ego}}(E_t), \mathbf{z}_t). \quad (4)$$

B_t is the BEV belief input at time t (the accumulated field L_t together with recently re-rendered instantaneous ray maps) and $f_{\text{BEV}}(\cdot)$ is the CNN encoder above that maps it to a pooled feature grid. D_t is the set of per-source DoA tokens at time t (bearing plus role information for every currently tracked source) and $f_{\text{DoA}}(\cdot)$ is the DoA encoder that aggregates them via masked mean/max pooling and temporal processing. E_t is the ego-state input at time t (recent pose history and the field's spread/confidence readouts) and $f_{\text{ego}}(\cdot)$ is its encoder. $\mathbf{z}_t$ is the pooled audio latent carried forward from the associated source's $\mathbf{z}_{t,i}$ (Sec. III-A). $f_{\text{fuse}}(\cdot)$ is the fusion layer that combines all four encoded streams into the single conditioning vector $\mathbf{c}_t$ used below.

For a verbally conditioned checkpoint, the Whisper phrase score and encoder embedding are additional inputs. The default policy does not receive newly enabled verbal features at deployment.

b) Diffusion training and inference.

The policy represents a normalized trajectory as $\tau_0 \in \mathbb{R}^{80 \times 2}$. Training perturbs an expert trajectory with Gaussian noise at diffusion timestep k :

$$\tau_k = \sqrt{\bar{\alpha}_k} \tau_0 + \sqrt{1 - \bar{\alpha}_k} \varepsilon, \quad \varepsilon \sim \mathcal{N}(\mathbf{0}, \mathbf{I}). \quad (5)$$

where τ_0 is the clean expert trajectory (80 waypoints in the plane) and k indexes the diffusion timestep, i.e. how much noise has been mixed in. $\bar{\alpha}_k \in (0, 1]$ is the standard DDPM cumulative noise-schedule coefficient at step k (the product of per-step signal-retention factors), so that $\bar{\alpha}_k$ decreases from near 1 at $k=0$ (clean trajectory) toward 0 as k grows (pure noise). ε is the Gaussian noise sample actually added, drawn once per training example from a standard normal $\mathcal{N}(\mathbf{0}, \mathbf{I})$, and τ_k is the resulting noised trajectory at step k that the network is trained to denoise.

A conditional 1-D U-Net uses feature-wise linear modulation (FiLM) and skip connections to predict the injected noise. The noise-prediction objective is:

$$\mathcal{L}_{\text{diff}} = \mathbb{E} \left[\|\varepsilon - \varepsilon_{\theta}(\tau_k, k, \mathbf{c}_t)\|_2^2 \right]. \quad (6)$$

where $\varepsilon_{\theta}(\tau_k, k, \mathbf{c}_t)$ is the U-Net's predicted noise, a function of the noised trajectory τ_k , the timestep k , and the conditioning vector $\mathbf{c}_t$, with θ denoting the network's trainable weights (FiLM-modulated by k and $\mathbf{c}_t$). $\mathbb{E}[\cdot]$ is the expectation over training trajectories, sampled timesteps k , and noise draws ε , and $\|\cdot\|_2^2$ is the squared Euclidean norm over the trajectory's 80 waypoints. $\mathcal{L}_{\text{diff}}$ is the resulting scalar training loss minimized over θ .

An auxiliary source-regression head provides a learned source-position estimate when source supervision is available. Navigation training uses synthetic expert trajectories; frozen recognition supplies conditioning without being updated by the diffusion loss.

At inference, generation starts from Gaussian noise and uses 16 DDIM iterations. The resulting 80 waypoints span 8 s at 10 Hz.

c) Post-sampling heading alignment.

The default online configuration refines trajectory direction using ODAS after denoising. Among eligible tracks with attraction posterior at least 0.05, it selects the bearing nearest the generated path heading within a 30° gate. If a track is selected, the entire trajectory is rotated about the robot by the heading difference:

$$\tilde{\mathbf{q}}_j = R(\Delta\theta) \mathbf{q}_j, \quad j = 1, \dots, 80. \quad (7)$$

$\mathbf{q}_j$ is the j -th of the 80 waypoints produced by DDIM sampling, in the robot frame. $\Delta\theta$ is the heading-correction angle: the difference between the selected ODAS track's bearing and the generated path's own heading, as selected above. $R(\Delta\theta)$ is the 2×2 rotation matrix by angle $\Delta\theta$ about the robot's origin, and $\tilde{\mathbf{q}}_j$ is the corresponding rotated waypoint; applying the same rotation to every j reorients the whole trajectory toward the selected source without changing its shape or speed profile.

IV. EXPERIMENTS AND RESULTS

A. Dataset Construction

Speech recordings are drawn from LibriSpeech [19], while emotional speech is obtained from RAVDESS [20]. Fearful speech serves as a proxy for distress; these acted utterances are not literal requests for assistance. Audio is processed in 10.24 s windows, with shorter recordings padded. Successive clips are concatenated without overlapping crossfades.

Synthetic scenes combine a designated caller, competing speakers, and non-speech distractors. A potential-field expert provides trajectory supervision. Simulated DoA observations are modeled on the detection patterns observed in real ODAS recordings collected using a microphone array mounted on our ZSL-1 quadruped. The simulation reproduces clutter detections, intermittent source tracks, and near-vertical phantom tracks. The controlled speech-first configuration uses negligible inter-source leakage. Thus, real recordings inform the acoustic observation model, while navigation trajectories and their supervision remain synthetic.

The evaluated validation partition contains 414 windows: 296 with a HELP-designated caller and 118 without a caller. Caller windows comprise 160 female-speech and 136

TABLE I: Human-directed navigation. Rates are mean $\pm$ sample standard deviation over three sampling seeds. N counts windows containing each class; classes may co-occur. Laughter is treated as a non-speech distractor.

Source class	N	Rate (%) $\uparrow$
Caller alignment within 45°		
Female speech	160	97.29 $\pm$ 1.30
Male speech	136	96.32 $\pm$ 0.74
Distractor rejection		
Domestic sounds	71	89.20 $\pm$ 0.81
Laughter	72	98.61 $\pm$ 0.00
Music	99	95.96 $\pm$ 0.00
Musical instrument	66	98.99 $\pm$ 0.87
Telephone	108	95.99 $\pm$ 1.07

male-speech examples; 112 additionally contain a competing human speaker. These labels describe the evaluation sources, rather than outputs of a trained multiclass recognition head. All navigation-policy ablations use this validation partition.

B. Human-Directed Navigation

We evaluate whether AcousticDiffusion follows callers while ignoring distracting sounds. Caller alignment measures end-point directions within 45° of the caller. Distractor rejection counts source–window pairs where the distractor is not both nearest in bearing to the end-point and within 20° of it. These metrics assess directional selection, not recognition accuracy or physical arrival.

Table I shows caller alignment above 96% and distractor rejection of 89.20–98.99%. With a competing speaker, the policy favors the HELP-designated caller in 91.07 $\pm$ 0.89% of windows, supporting learned human-directed behavior. Without a caller, mean seed-wise median end-point displacement falls from 2.86 to 0.27 m, near the expert’s 0.24 m; however, only 1.69% of predictions remain within 0.05 m, indicating reduced motion rather than reliable abstention. Improved emotion recognition could further strengthen caller prioritization; the current system uses distress as a preference, not a prerequisite for approaching a person.

C. Qualitative Real-Robot Results

Fig. 3 presents on-robot visualizations of AcousticDiffusion. The displayed trajectories and acoustic estimates illustrate the integration of microphone-array observations, Bayesian spatial belief, and learned trajectory generation within the ROS 2 system. The learned source-head and BEV estimates can differ, highlighting the distinction between the two localization outputs. These examples demonstrate on-robot operation and successful sound source classification and localization.

D. Ablation: Diffusion Prediction Target

We compare noise prediction (ϵ) and direct trajectory prediction ($\mathbf{x}_0$) with identical data, architecture, conditioning, and 100-epoch training. The recognizer remains frozen; evaluation uses final EMA checkpoints and 16-step DDIM without ODAS steering.

TABLE II: Prediction-target ablation. Alignment values are mean $\pm$ sample standard deviation over three sampling seeds, not training runs. ADE, FDE, and roughness use 256 validation windows.

Metric	$\mathbf{x}_0$	ϵ
Bearing error (°) $\downarrow$	17.42 $\pm$ 0.07	11.20 $\pm$ 0.33
Caller-directed (%) $\uparrow$	87.05 $\pm$ 0.39	89.64 $\pm$ 0.98
Within 30° (%) $\uparrow$	80.86 $\pm$ 0.20	91.78 $\pm$ 1.67
ADE (m) $\downarrow$	0.285	0.329
FDE (m) $\downarrow$	0.846	0.725
Roughness / expert $\downarrow$	136.29	76.98

TABLE III: Conditioning ablation: mean $\pm$ sample standard deviation over three sampling seeds. Bearing denotes end-point angular error to the caller.

Conditioning	Bearing (°) $\downarrow$	Within 30° (%) $\uparrow$
Full	11.20 $\pm$ 0.33	91.78 $\pm$ 1.67
No speech (prior)	20.03 $\pm$ 0.43	79.84 $\pm$ 0.70
No speech (zero)	24.16 $\pm$ 0.76	73.25 $\pm$ 1.66
No ODAS (prior)	46.76 $\pm$ 1.30	32.58 $\pm$ 0.97
No ODAS (zero)	24.16 $\pm$ 0.85	79.69 $\pm$ 2.09
Past motion only	48.27 $\pm$ 1.04	25.14 $\pm$ 2.16
Individual component removal		
Token roles	11.27 $\pm$ 0.28	90.77 $\pm$ 1.67
Recognizer latent	10.55 $\pm$ 0.04	92.23 $\pm$ 0.68
BEV role maps	19.44 $\pm$ 0.45	81.06 $\pm$ 1.21
Field role readout	11.65 $\pm$ 0.40	91.33 $\pm$ 1.86

Bearing error measures the angular difference between end-point and caller directions. Caller-directed rate identifies the caller as the nearest source in bearing; within-30° measures alignment within that tolerance. Neither implies physical arrival. ADE and FDE measure mean waypoint and final-point Euclidean errors against the expert, respectively. Relative roughness divides predicted roughness by expert roughness, with one indicating equal roughness.

Table II reveals a trade-off between imitation accuracy and acoustic alignment: $\mathbf{x}_0$ prediction achieves lower ADE, whereas ϵ prediction reduces bearing error by 35.7%, improves within-30° alignment by 10.92 percentage points, and lowers FDE. It also reduces roughness by 43.5%, although both variants remain substantially rougher than the expert. We therefore adopt ϵ prediction for its better target alignment and end-point accuracy; dependence on individual conditioning inputs is examined separately.

E. Ablation: Conditioning Dependence

We remove conditioning inputs at inference time from a fixed ϵ -prediction checkpoint, replacing them with unobserved priors or zeros. Speech removal affects token roles, the recognizer latent, BEV role maps, and the field role readout; ODAS removal affects DoA tokens, all BEV maps, and the field estimate. This tests input dependence without retraining.

Removing speech or ODAS conditioning degrades alignment under both replacement schemes, while past motion alone reduces within-30° alignment from 91.78% to 25.14%. Among individual semantic components, BEV role-map removal causes the largest degradation, identifying spatial

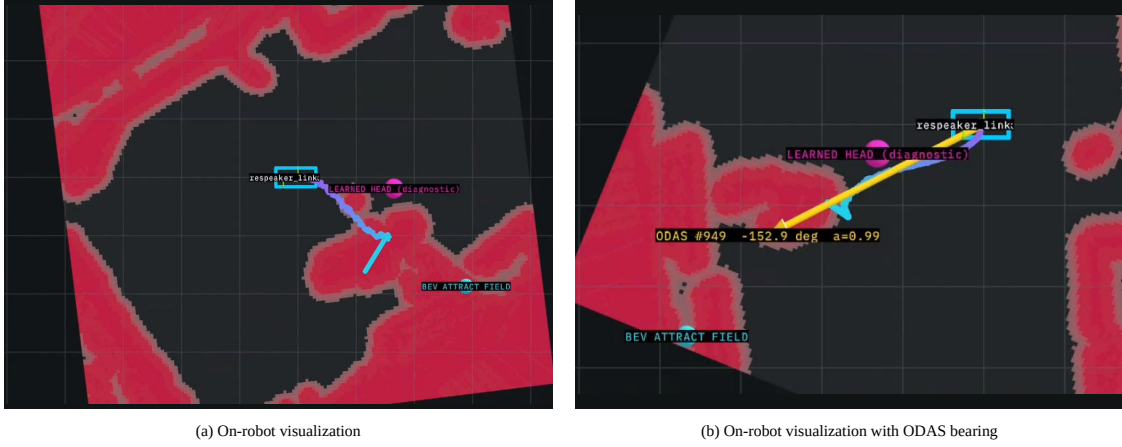


Fig. 3: Qualitative visualizations from the AcousticDiffusion deployment on the quadruped robot. Both panels show the microphone-array frame (`respeaker_link`), the displayed trajectory, the learned source-head diagnostic (magenta), and the BEV attraction-field estimate (red), overlaid on the local map. Panel (b) additionally shows the ODAS bearing observation in yellow.

semantics as the principal tested semantic pathway. Removing the recognizer latent slightly improves alignment but increases relative roughness from 77.59 to 102.23. We retain full conditioning for quadruped experiments.

V. REAL-ROBOT EVALUATION

A. Robot Setup

Experiments used a ZSL-1 quadruped equipped with an NVIDIA Jetson Orin NX, a Livox MID-360 LiDAR/IMU, and a 16-kHz Saeed reSpeaker XVF3800 four-microphone array, with ODAS 2.1 supplying direction-of-arrival tracks and separated audio. The array centre sits 0.25 m behind and 0.30 m above `base_link`, rotated 180° in yaw. All perception, inference, and navigation ran on-board, with no external compute.

Localization fused motor odometry with LiDAR-IMU measurements via planar NDT-OMP scan matching against a prior point-cloud map. Navigation used LiDAR costmaps with Nav2’s MPPI controller, capped at 0.35 m/s linear and 0.50 rad/s angular speed.

An external Vicon system, tracking the microphone array and multiple acoustic emitters, supplied ground truth for evaluation only. Vicon and map coordinates were aligned per run; since emitter markers had zero horizontal offset from the physical sources, only height offsets were applied.

B. Source Separation Configuration

Microphone geometry was verified against manufacturer coordinates, with approximately 6.6 cm inter-microphone spacing. We evaluated delay-and-sum (DS) beamforming and geometric source separation with decorrelation (DGSS), each with and without ODAS spectral-masking post-filtering to suppress residual noise and inter-source interference. ODAS’s Kalman filter tracked source directions over time, separately from the audio post-filter.

Each configuration was evaluated with two active speakers using mean pairwise correlation between active output channels as a proxy for cross-talk. DS and DGSS required

separate recording sessions and produced two and three active output channels, respectively, due to dynamic slot assignment, limiting direct comparison.

Post-filtering reduced correlation from 0.393 to 0.001 for DS and from 0.376 to 0.061 for DGSS. Without post-filtering, the methods produced similar correlations. We adopt DGSS with post-filtering, which reduced correlation by approximately 84% relative to DGSS alone, although DS with post-filtering achieved the lowest measured correlation.

C. ODAS-Coupled Baselines

A* and RRT* use the same active ODAS SST track identifier logged by AcousticDiffusion, avoiding independent source selection. At each replanning cycle, the robot origin serves as the start, and the track’s reported robot-frame Cartesian coordinates define the goal. A* searches an 8-connected grid at 0.10 m resolution with a Euclidean heuristic; RRT* uses nearest-neighbor extension and cost-based rewiring within a bounded local region. Replanning is requested at approximately 1.5 Hz, independently of the recognizer’s 10.24 s analysis window.

D. Comparison with Classical Planners

Table IV compares AcousticDiffusion with A* and RRT on trials S1–S3, using the same acoustic bearing estimates and costmaps. We report instantaneous bearing error and the minimum bearing error within each 10.24 s window, averaged over windows. The window duration matches the audio recognizer’s analysis interval, allowing assessment over the same temporal horizon used for semantic conditioning. The windowed metric measures whether accurate alignment occurs during that interval, rather than whether it is maintained throughout. We additionally report the fraction of windows containing an alignment error of at most 15°.

AcousticDiffusion achieves the lowest instantaneous bearing error (64.9°) and windowed error (47.2°), compared with 98.2° and 72.8° for A*, and 90.4° and 59.8° for RRT. It reaches 15° alignment in 32% of windows, versus

TABLE IV: Comparison on trials S1–S3. AD denotes AcousticDiffusion. Windowed error is the mean of the minimum bearing errors within 10.24 s windows. Compute time measures planner execution, not end-to-end sensing latency. Bold indicates the best value, excluding the descriptive replanning period.

Metric	AD (ours)	A*	RRT
<i>Bearing alignment</i>			
Mean error ($^{\circ}$) ↓	64.9	98.2	90.4
Windowed error ($^{\circ}$) ↓	47.2	72.8	59.8
Windows $\leq 15^{\circ}$ (%) ↑	32	18	25
<i>Planner compute and timing</i>			
Mean compute (ms) ↓	6.07	0.15	23.6
P95 compute (ms) ↓	12.67	0.23	25.2
Median replan period (ms)	645.7	682.1	682.1
<i>Final source distance</i>			
Mean (m) ↓	2.48	3.96	3.96
Median (m) ↓	2.48	3.42	3.42

18% and 25%, respectively. These results indicate improved directional guidance, although the remaining errors show that accurate alignment is not consistently maintained.

AcousticDiffusion’s mean planner compute time is 6.07 ms, faster than RRT’s 23.6 ms but slower than A*’s 0.15 ms on this costmap. Its P95 compute time of 12.67 ms remains well below the observed median replanning period of roughly 650 ms, as do the compute times of both baselines, so none of the three planners is compute bound at the replanning rate evaluated here. Finally, the reported mean final source distance decreases from 3.96 to 2.48 m, a 37.4% reduction. Together, these measurements favor AcousticDiffusion in directional alignment and final proximity within this evaluation.

VI. CONCLUSION AND FUTURE WORK

We presented AcousticDiffusion, a speech-guided navigation framework combining pretrained audio recognition, a semantic Bayesian BEV sound field, and diffusion-based trajectory generation. On the validation set, the policy achieves a mean end-point bearing error of 11.20° , with 91.78% of trajectories aligned within 30° of the caller. Distractor rejection ranges from 89.20% to 98.99%, and the HELP-designated caller is favored over a competing speaker in 91.07% of windows. Deployed online on the quadruped without additional retraining, the policy achieves a mean bearing error of 64.9° , compared with 98.2° for A* and 90.4° for RRT, with a mean planner compute time of 6.07 ms. The larger real-robot error highlights the remaining simulation-to-reality gap.

Future work will fuse acoustic and visual features to improve localization and obstacle-aware navigation: audio guides motion toward occluded callers, while vision supplies spatial context. Conditional diffusion can integrate these complementary cues and represent multiple plausible trajectories under uncertainty. We will investigate whether this fusion can reduce reliance on dedicated depth sensing and explicit maps by learning collision-aware trajectories directly from observations.

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